Year II — Mathematical Methods

Based on lectures by Various Lecturers

Notes taken by Anonymous

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Chapter 1

Derivatives and Coordinates

1.1 Vector and Scalar Functions

1.1.1 Vector Functions

Definition 1.1.1 (Vector Function)**.** A *vector function* f is a function of the form:

$$f : \mathbb{R} \to \mathbb{R}^n$$

which takes a numerical input and returns a vector

Definition 1.1.2 (Derivative of a Vector Function)**.** A vector function $\mathbf{F}(x)$ is differentiable if

$$\delta\mathbf{F} \overset{\text{def}}{=} \mathbf{F}(x+\delta x) - \mathbf{F}(x) = \mathbf{F}'(x)\delta x + o(\delta x)$$

for some $\mathbf{F}'(x)$.$\mathbf{F}'(x)$ is called the derivative of $\mathbf{F}(x)$. This is the equivalent of writing:

$$\mathbf{F}' = \frac{\mathrm{d}\mathbf{F}}{\mathrm{d}x} = \lim_{\delta x \to 0} \frac{1}{\delta x}[\mathbf{F}(x+\delta x) - \mathbf{F}(x)]$$

Example 1.1.3.

$$\mathbf{r}(t) = \begin{pmatrix} t^2 \\ e^t \\ \sin t \end{pmatrix}$$

Differentiate componentwise:

$$\mathbf{r}'(t) = \begin{pmatrix} 2t \\ e^t \\ \cos t \end{pmatrix}$$

At $t = 0$:

$$\mathbf{r}'(0) = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

Proposition 1.1.4 (Componentwise differentiation)**.** Using differential notation, the differentiability condition can be written as

$$\mathrm{d}\mathbf{F} = \mathbf{F}'(x)\mathrm{d}x$$

Given a basis $\mathbf{e}_i$ that is independent of x, vector differentiation is performed component wise:

$$\mathbf{F}'(x) = F_i'(x)\mathbf{e}_i.$$

Proposition 1.1.5 (Vector Differentiation on Operations)**.**

$$\frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{fg}) = \frac{\mathrm{d}\mathbf{f}}{\mathrm{d}t}\mathbf{g} + \mathbf{f}\frac{\mathrm{d}\mathbf{g}}{\mathrm{d}t}$$
$$\frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{g}\cdot\mathbf{h}) = \frac{\mathrm{d}\mathbf{g}}{\mathrm{d}t}\cdot\mathbf{h} + \mathbf{g}\cdot\frac{\mathrm{d}\mathbf{h}}{\mathrm{d}t}$$
$$\frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{g}\times\mathbf{h}) = \frac{\mathrm{d}\mathbf{g}}{\mathrm{d}t}\times\mathbf{h} + \mathbf{g}\times\frac{\mathrm{d}\mathbf{h}}{\mathrm{d}t}$$

We note here that the order of operations matters for the case of the dot product since it is anti symmetric

1.1.2 Scalar Fields

Definition 1.1.6 (Scalar Field)**.** A *scalar field* f is a mapping of the form:

$$f : \mathbb{R}^n \to \mathbb{R}$$

which takes a vector and returns a numerical output.

Remark 1.1.7 (Distinction between Vector and Scalar Function)**.** A scalar function takes a position and returns a numerical output (example : potential at a point) where a vector function takes a number and maps it to a vector (example : the velocity associated to a particle at t)

Definition 1.1.8 (Limit of a Vector)**.** Definition (Limit of vector). The limit of vectors is defined using the norm. So $\mathbf{v} \to \mathbf{c}$ iff $|\mathbf{v} - \mathbf{c}| \to 0$. So a vector approaches another vector if and only if the difference between them goes to 0. Similarly, $f(\mathbf{r}) = o(\mathbf{r})$ means $\frac{|f(\mathbf{r})|}{|\mathbf{r}|} \to 0$ as $\mathbf{r} \to \mathbf{0}$.

Definition 1.1.9 (Gradient / Derivative of a Scalar Function)**.** A scalar function $f(\mathbf{r})$ is differentiable at $\mathbf{r}$ if

$$\delta f \overset{\text{def}}{=} f(\mathbf{r}+\delta\mathbf{r}) - f(\mathbf{r}) = (\nabla f)\cdot\delta\mathbf{r} + o(\delta\mathbf{r})$$

for some vector ∇f, the gradient of f at $\mathbf{r}$.

Example 1.1.10.

$$f(x,y) = x^2 y + \sin y$$

The derivative is the gradient row/column (depending on convention). The partial derivatives are: - $\frac{\partial f}{\partial x} = 2xy$ - $\frac{\partial f}{\partial y} = x^2 + \cos y$
Gradient:

$$\nabla f(x,y) = \begin{pmatrix} 2xy \\ x^2 + \cos y \end{pmatrix}$$

At $(1,0)$:

$$\nabla f(1,0) = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

Remark 1.1.11 (Importance of Taylor Derivative Definition)**.** We see here the importance of this new way of defining a derivative in higher dimensions since dividing by a vector makes no sense if we were to try and rearrange.

Definition 1.1.12 (Directional Derivative)**.** The directional derivative of f along $\mathbf{n}$ is

$$\mathbf{n} \cdot \nabla f = \lim_{h \to 0} \frac{1}{h}[f(\mathbf{r} + h\mathbf{n}) - f(\mathbf{r})],$$

It refers to how fast f changes when we move in the direction of $\mathbf{n}$.

Remark 1.1.13 (Maximising the Directional Derivative)**.** When the derivative moves in the direction of $\propto$ then we have the maximum possible value of the directional derivative and thus points in the direction of the greatest slope

Theorem 1.1.14 (Gradient expressed in Partial Derivative)**.** The gradient is

$$\nabla f = \frac{\partial f}{\partial x_i}\mathbf{e}_i$$

Hence we can write the condition of differentiability as

$$\delta f = \frac{\partial f}{\partial x_i}\delta x_i + o(\delta \mathbf{x}).$$

In differential notation, we write

$$\mathrm{d}f = \nabla f \cdot \mathrm{d}\mathbf{r} = \frac{\partial f}{\partial x_i}\,\mathrm{d}x_i,$$

which is the chain rule for partial derivatives.

Theorem 1.1.15 (Chain Rule for a Scalar Function)**.** Given a function $f(\mathbf{r}(u))$,

$$\frac{\mathrm{d}f}{\mathrm{d}u} = \nabla f \cdot \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}u} = \frac{\partial f}{\partial x_i}\frac{\mathrm{d}x_i}{\mathrm{d}u}.$$

Note that if we drop the du, we simply get

$$\mathrm{d}f = \nabla f \cdot \mathrm{d}\mathbf{r} = \frac{\partial f}{\partial x_i}\,\mathrm{d}x_i,$$

which is what we've previously had.

1.1.3 Vector Fields

Definition 1.1.16 (Vector Field)**.** A *vector field* f is a function of the form:

$$f : \mathbb{R}^n \to \mathbb{R}^m$$

which takes a vector input of dimension n and returns a vector of dimension m

Definition 1.1.17 (Derivative of a Vector Field)**.** A vector field $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable if

$$\delta\mathbf{F} \overset{\text{def}}{=} \mathbf{F}(\mathbf{x}+\delta\mathbf{x}) - \mathbf{F}(\mathbf{x}) = M\delta\mathbf{x} + o(\delta\mathbf{x})$$

for some $m \times n$ matrix M.M is the derivative of $\mathbf{F}$.

Theorem 1.1.18 (Derivative of a Vector field in ij notation)**.** Given an arbitrary function $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^m$ that maps $\mathbf{x} \mapsto \mathbf{y}$ and a choice of basis, we can write $\mathbf{F}$ as a set of m functions $y_j = F_j(\mathbf{x})$ such that $\mathbf{y} = (y_1, y_2, \cdots, y_m)$. Then

$$\mathrm{d}y_j = \frac{\partial F_j}{\partial x_i}\ \mathrm{d}x_i.$$

and we can write the derivative as

$$M_{ji} = \frac{\partial y_j}{\partial x_i}.$$

Note that we could have used this as the definition of the derivative. However, the original definition is superior because it does not require a selection of coordinate system.

Example 1.1.19. Take the vector field

$$F(x,y) = \begin{pmatrix} x^2 y \\ \sin(xy) \end{pmatrix}.$$

The derivative matrix is obtained by differentiating each component with respect to each variable.

$$DF(x,y) = \begin{pmatrix} \frac{\partial}{\partial x}\left(x^2 y\right) & \frac{\partial}{\partial y}\left(x^2 y\right) \\ \frac{\partial}{\partial x}(\sin(xy)) & \frac{\partial}{\partial y}(\sin(xy)) \end{pmatrix}.$$

Computing the entries: - $\frac{\partial}{\partial x}\left(x^2 y\right) = 2xy$ - $\frac{\partial}{\partial y}\left(x^2 y\right) = x^2$ - $\frac{\partial}{\partial x}(\sin(xy)) = y\cos(xy)$ - $\frac{\partial}{\partial y}(\sin(xy)) = x\cos(xy)$

So

$$DF(x,y) = \begin{pmatrix} 2xy & x^2 \\ y\cos(xy) & x\cos(xy) \end{pmatrix}$$

Example at the point $(1,2)$:

$$DF(1,2) = \begin{pmatrix} 4 & 1 \\ 2\cos(2) & \cos(2) \end{pmatrix}.$$

Definition 1.1.20 (Smooth Function)**.** A function is smooth if it can be differentiated any number of times. This requires that all partial derivatives exist and are totally symmetric in i, j and k (i.e. the differential operator is commutative).

Theorem 1.1.21 (Chain Rule of Vector Fields)**.** Suppose $g : \mathbb{R}^p \to \mathbb{R}^n$ and $f : \mathbb{R}^n \to \mathbb{R}^m$. Suppose that the coordinates of the vectors in $\mathbb{R}^p, \mathbb{R}^n$ and $\mathbb{R}^m$ are u_a, x_i and y_r respectively. By the chain rule,

$$\frac{\partial y_r}{\partial u_a} = \frac{\partial y_r}{\partial x_i}\frac{\partial x_i}{\partial u_a},$$

with summation implied. Writing in matrix form,

$$M(f \circ g)_{ra} = M(f)_{ri} M(g)_{ia}.$$

Alternatively, in operator form,

$$\frac{\partial}{\partial u_a} = \frac{\partial x_i}{\partial u_a}\frac{\partial}{\partial x_i}.$$

1.2 Inverse Functions

Theorem 1.2.1 (Derivative of an Inverse Function in Vector Field)**.** Suppose $g, f : \mathbb{R}^n \to \mathbb{R}^n$ are inverse functions, i.e. $g \circ f = f \circ g = \mathrm{id(x)}$. This is where $\mathrm{id}(\mathbf{x}) = \mathbf{x}$ Suppose that $f(\mathbf{x}) = \mathbf{u}$ and $g(\mathbf{u}) = \mathbf{x}$.

Since the derivative of the identity function $\mathrm{id}(\mathbf{x})$ is the identity matrix (if you differentiate $\mathbf{x}$ wrt to $\mathbf{x}$, you get 1), we must have

$$M(f \circ g) = I.$$

Therefore we know that

$$M(g) = M(f)^{-1}.$$

We derive this result more formally by noting

$$\frac{\partial u_b}{\partial u_a} = \delta_{ab}.$$

So by the chain rule,

$$\frac{\partial u_b}{\partial x_i}\frac{\partial x_i}{\partial u_a} = \delta_{ab},$$

i.e. $M(f \circ g) = I$.

Remark 1.2.2 (Determinants)**.** For the inverse functions f and g:

$$\det M(f) \det M(g) = 1$$

1.3 Coordinate Systems

Cylindrical polars	**Spherical polars**
Conversion formulae	
$x_1 = \rho\cos\varphi$ $x_2 = \rho\sin\varphi$ $x_3 = z$	$x_1 = r\sin\theta\cos\varphi$ $x_2 = r\sin\theta\sin\varphi$ $x_3 = r\cos\theta$
Basis vectors	
$\mathbf{e}_\rho = (\cos\varphi, \sin\varphi, 0)$ $\mathbf{e}_\varphi = (-\sin\varphi, \cos\varphi, 0)$ $\mathbf{e}_z = (0, 0, 1)$	$\mathbf{e}_r = (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)$ $\mathbf{e}_\varphi = (-\sin\varphi, \cos\varphi, 0)$ $\mathbf{e}_\theta = (\cos\theta\cos\varphi, \cos\theta\sin\varphi, -\sin\theta)$

Chapter 2

Line Integrals

2.1 Parametrised curves, lengths and arc length

Definition 2.1.1 (Parametrisation)**.** Given a curve C in $\mathbb{R}^n$, a *parametrisation* of it is a continuous and invertible function $\mathbf{r} : D \to \mathbb{R}^n$ for some $D \subseteq \mathbb{R}$ whose image is C. Parametrisation is therefore the function mapping a real input into a vector output.

Definition 2.1.2 (Differentiable Parametrisation)**.** A parametrisation $\boldsymbol{x}(t)$ is differentiable if it has a well-defined tangent vector,

$$\frac{d\boldsymbol{x}}{dt} = \lim_{dt\to 0} \frac{\boldsymbol{x}(t+dt) - \boldsymbol{x}(t)}{dt},$$

so-called because its direction points along the curve at each t.

Definition 2.1.3 (Regular Parametrisation)**.** A parametrisation is regular if $\frac{d\boldsymbol{x}}{dt} \neq 0$ everywhere, and a curve is regular if it has at least one regular parametrisation.

Definition 2.1.4 (Singular Parametrisation and Cusps)**.** A parametrisation is singular if $\frac{d\boldsymbol{x}}{dt} = 0$ at at least one point. The singular points of a parametrisation are called susps.

Remark 2.1.5. You can construct a singular parametrisation even for a regular curve.

Proposition 2.1.6 (Change in Arclength of a curve)**.** Let s denote the arclength of a curve $\mathbf{r}(u)$. Then

$$\frac{\mathrm{d}s}{\mathrm{d}u} = \pm\left|\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}u}\right| = \pm\sqrt{\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}u}\cdot\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}u}} = \pm\left|\mathbf{r}'(u)\right|$$

with the sign depending on whether it is in the direction of increasing or decreasing arclength.

Remark 2.1.7 (Arclength)**.** Using the formula above, the arc length between two points on the curve $\mathbf{r}(u)$, given by $u = u_1$ and $u = u_2$, is

$$s = \int_{u_1}^{u_2} \sqrt{\frac{d\mathbf{r}}{du}\cdot\frac{d\mathbf{r}}{du}}\,du \tag{2.1}$$

Method 2.1.8 (Deriving tiny change in Arclength)**.** The tiny displacement vector is

$$d\mathbf{r} = \mathbf{r}'(u)du$$

Its magnitude gives the tiny distance travelled:

$$ds = |d\mathbf{r}|$$

Hence

$$ds = |\mathbf{r}'(u)du|$$

and since du is scalar,

$$ds = |\mathbf{r}'(u)|\,|du|$$

which becomes

$$ds = |\mathbf{r}'(u)|\,du$$

when moving in the positive parameter direction.

2.2 Line Integrals of Scalar Fields

Definition 2.2.1 (Scalar Line Element)**.** The scalar line element of C is $\mathrm{d}s$.

Proposition 2.2.2 (Formula for Scalar Line Element)**.** $\mathrm{d}s = \pm\,|\mathbf{r}'(u)|\,\mathrm{d}u$

Definition 2.2.3 (Line Integral of a Scalar Field)**.** The line integral of a scalar field $f(\boldsymbol{x})$ along a curve C with parametrisation $\boldsymbol{x}(t)$, for $t \in [t_0, t_1]$, is defined as

$$\int_C f\,ds = \int_{t_0}^{t_1} f(\boldsymbol{x}(t)) \left|\frac{d\boldsymbol{x}}{dt}\right| dt.$$

Proposition 2.2.4 (*Choice of Parametrisation doesn't matter)**.** The line integral of a scalar field is independent of the choice of parametrisation.

Remark 2.2.5. The arclength formula from before is therefore the line integral of the scalar field $f(\mathbf{x}) = 1$

Definition 2.2.6 (Rectifiable)**.** A curve which has finite length is called rectifiable. This is a weaker condition than "regular" (for example, if $\boldsymbol{x}(t)$ is Lipschitz continuous then it is rectifiable).

2.3 Line Integral of Vector Fields

Definition 2.3.1 (Line Integral of a Vector Field)**.** The *line integral* of a smooth vector field $\mathbf{F}(\mathbf{r})$ along a path C parametrised by $\mathbf{r}(u)$ along the direction (orientation) $\mathbf{r}(\alpha) \to \mathbf{r}(\beta)$ is

$$\int_C \mathbf{F}(\mathbf{r}) \cdot \mathrm{d}\mathbf{r} = \int_\alpha^\beta \mathbf{F}(\mathbf{r}(u)) \cdot \mathbf{r}'(u)\mathrm{d}u$$

We say $\mathrm{d}\mathbf{r} = \mathbf{r}'(u)\mathrm{d}u$ is the line element on C. Note that the upper and lower limits of the integral are the end point and start point respectively, and β is not necessarily larger than α.

Example 2.3.2 (Showing Line Integral depends on Curve)**.** Take $\mathbf{F}(\mathbf{r}) = \left(xe^y, z^2, xy\right)$ and we want to find the line integral from $\mathbf{a} = (0,0,0)$ to $\mathbf{b} = (1,1,1)$.

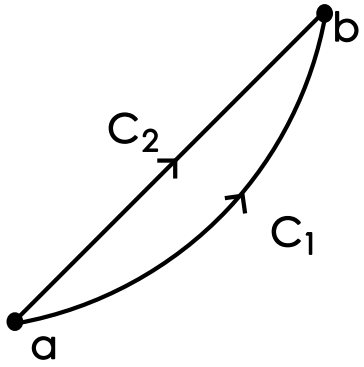


Figure 2.1: Two curves between a and b

We first integrate along the curve $C_1 : \mathbf{r}(u) = \left(u, u^2, u^3\right)$. Then $\mathbf{r}'(u) = \left(1, 2u, 3u^2\right)$, and $\mathbf{F}(\mathbf{r}(u)) = \left(ue^{u^2}, u^6, u^3\right)$. So

$$\begin{aligned}
\int_{C_1} \mathbf{F} \cdot \ \mathrm{d}\mathbf{r} &= \int_0^1 \mathbf{F} \cdot \mathbf{r}'(u)\mathrm{d}u \\
&= \int_0^1 ue^{u^2} + 2u^7 + 3u^5 \ \mathrm{d}u \\
&= \frac{e}{2} - \frac{1}{2} + \frac{1}{4} + \frac{1}{2} \\
&= \frac{e}{2} + \frac{1}{4}
\end{aligned}$$

Now we try to integrate along another curve $C_2 : \mathbf{r}(t) = (t,t,t)$. So $\mathbf{r}'(t) = (1,1,1)$.

$$\begin{aligned}
\int_{C_2} \mathbf{F} \cdot \ \mathrm{d}\mathbf{r} &= \int \mathbf{F} \cdot \mathbf{r}'(t)\mathrm{d}t \\
&= \int_0^1 te^t + 2t^2 \ \mathrm{d}t \\
&= \frac{5}{3}
\end{aligned}$$

We see that the line integral depends on the curve C in general, not just $\mathbf{a}, \mathbf{b}$.

Definition 2.3.3 (Unit Tangent)**.** The unit tangent vector of a parametrisation is

$$\hat{\boldsymbol{t}} = \frac{\frac{d\boldsymbol{x}}{dt}}{\left|\frac{d\boldsymbol{x}}{dt}\right|}$$

This only depends on the direction of the parametrisation, not the "speed".

Definition 2.3.4 (Oriented Curve)**.** An oriented curve is a curve together with a specified (consistent) choice of $\hat{\boldsymbol{t}}$. We indicate this by adding an arrow to the curve.

Definition 2.3.5 (Line Integral along an Oriented Curve)**.** The line integral of a vector field $\boldsymbol{f}(\boldsymbol{x})$ along an oriented curve C with parametrisation $\boldsymbol{x}(t)$, for $t \in [t_0, t_1]$, is defined as the scalar line integral of the tangential component:

$$\int_C \boldsymbol{f} \cdot \hat{\boldsymbol{t}} d\ell = \int_{t_0}^{t_1} \boldsymbol{f}(\boldsymbol{x}(t)) \cdot \hat{\boldsymbol{t}} \left| \frac{d\boldsymbol{x}}{dt} \right| dt = \int_{t_0}^{t_1} \boldsymbol{f}(\boldsymbol{x}(t)) \cdot \frac{d\boldsymbol{x}}{dt} dt$$

Remark 2.3.6. If the parametrisation does not trace the curve, the integral sign will flip

Definition 2.3.7 (Closed Curve)**.** A closed curve is a curve with the same start and end point. The line integral along a closed curve is (sometimes) written as $\oint$ and is (sometimes) called the circulation of $\mathbf{F}$ around C.

Remark 2.3.8 (Circulation using Oriented Curve)**.** If the curve C is closed, this is often emphasised by the notation

$$\int_C \boldsymbol{f} \cdot \hat{\boldsymbol{t}} d\ell = \oint_C \boldsymbol{f} \cdot \hat{\boldsymbol{t}} d\ell$$

and the line integral is called the circulation of $\boldsymbol{f}$ around C.

Definition 2.3.9 (Piecewise Smooth Curve)**.** A piecewise smooth curve is a curve $C = C_1 + C_2 + \cdots + C_n$ with all C_i smooth with regular parametrisations. The line integral over a piecewise smooth C is

$$\int_C \mathbf{F} \cdot \, \mathrm{d}\mathbf{r} = \int_{C_1} \mathbf{F} \cdot \, \mathrm{d}\mathbf{r} + \int_{C_2} \mathbf{F} \cdot \, \mathrm{d}\mathbf{r} + \cdots + \int_{C_n} \mathbf{F} \cdot \, \mathrm{d}\mathbf{r}.$$

Example 2.3.10 (Turning two curves into piecewise smooth closed curve)**.** Take the example above, and let $C_3 = -C_2$. Then $C = C_1 + C_3$ is piecewise smooth but not smooth:

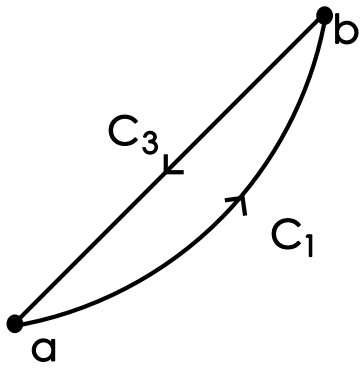


Figure 2.2: Closed line integral between a and b

Then

$$\begin{aligned}\oint_C \mathbf{F}\cdot\ \mathrm{d}\mathbf{r} &= \int_{C_1}\mathbf{F}\cdot\ \mathrm{d}\mathbf{r} + \int_{C_3}\mathbf{F}\cdot\ \mathrm{d}\mathbf{r}\\ &= \left(\frac{e}{2}+\frac{1}{4}\right)-\frac{5}{3}\\ &= -\frac{17}{12}+\frac{e}{2}\end{aligned}$$

2.4 Gradients and Differentials

Theorem 2.4.1 (*Line Integrals which depend on only endpoints)**.** If $\mathbf{F}=\nabla f(\mathbf{r})$, then

$$\int_C \mathbf{F}\cdot\ \mathrm{d}\mathbf{r} = f(\mathbf{b})-f(\mathbf{a}),$$

where $\mathbf{b}$ and $\mathbf{a}$ are the end points of the curve. Therefore the integral depends only on the end points and not the curve itself.

Definition 2.4.2 (Conservative Vector Field)**.** If $\mathbf{F}=\nabla f$ for some f, the $\mathbf{F}$ is called a conservative vector field.

Remark 2.4.3. 'Conservative' comes from mechanics where energy is conserved in a system. The work done around a closed curve is 0.

Definition 2.4.4 (Exact Differential)**.** A differential $\mathbf{F}\cdot\mathrm{d}\mathbf{r}$ is exact if there is an f such that $\mathbf{F}=\nabla f$. Then

$$\mathrm{d}f = \nabla f\cdot\ \mathrm{d}\mathbf{r} = \frac{\partial f}{\partial x_i}\ \mathrm{d}x_i.$$

Proposition 2.4.5 (Test for Conservative Vector Field)**.** If $\mathbf{F}=\nabla f$ for some f, then

$$\frac{\partial F_i}{\partial x_j} = \frac{\partial F_j}{\partial x_i}$$

This is because both are equal to $\partial^2 f/\partial x_i\partial x_j$. For an exact differential, the result from the previous section reads

$$\int_C \mathbf{F}\cdot\ \mathrm{d}\mathbf{r} = \int_C \mathrm{d}f = f(\mathbf{b})-f(\mathbf{a})$$

Proposition 2.4.6 (Manipulations on differentials)**.**

$$\begin{aligned}\mathrm{d}(\lambda f+\mu g) &= \lambda\mathrm{d}f+\mu\mathrm{d}g\\ \mathrm{d}(fg) &= (\mathrm{d}f)g+f(\ \mathrm{d}g)\end{aligned}$$

Using these, it may be possible to find f by inspection. We do this by integrating each component and seeing if they are equivalent. If they are, then the line integral is just this function evaluated between $\mathbf{a}$ and $\mathbf{b}$.

2.5 Work and Potential Energy

Definition 2.5.1 (Work Done). If $\mathbf{F}(\mathbf{r})$ is a force, then $\int_C \mathbf{F} \cdot \, \mathrm{d}\mathbf{r}$ is the work done by the force along the curve C. It is the limit of a sum of terms $\mathbf{F}(\mathbf{r}) \cdot \delta\mathbf{r}$, i.e. the force along the direction of $\delta\mathbf{r}$.

Definition 2.5.2 (Potential Energy). Given a conservative force $\mathbf{F} = -\nabla V, V(\mathbf{x})$ is the potential energy. Then

$$\int_C \mathbf{F} \cdot \, \mathrm{d}\mathbf{r} = V(\mathbf{a}) - V(\mathbf{b}).$$

Therefore, for a conservative force, we have $\mathbf{F} = \nabla V$, where $V(\mathbf{r})$ is the potential energy.

2.6 Curvilinear Coordinates

Definition 2.6.1 (Curvilinear Coordinates). In a curvilinear coordinate system (u, v, w), the coordinate lines - where two of the coordinates are constant and one varies - may be curved. These coordinate lines may be parametrised by the corresponding coordinate, so are given by $\boldsymbol{x}(u, v, w)$ with two of the coordinates fixed. The corresponding tangent vectors are therefore $\frac{\partial \boldsymbol{x}}{\partial u}, \frac{\partial \boldsymbol{x}}{\partial v}, \frac{\partial \boldsymbol{x}}{\partial w}$.

Definition 2.6.2 (Orthogonal and Scale Factors). The coordinate system is orthogonal if the three tangent vectors are mutually orthogonal at every point. We can then define an orthonormal basis $\{\boldsymbol{e}_u, \boldsymbol{e}_v, \boldsymbol{e}_w\}$ by normalising:

$$\boldsymbol{e}_u = \frac{1}{h_u}\frac{\partial \boldsymbol{x}}{\partial u}, \quad \boldsymbol{e}_v = \frac{1}{h_v}\frac{\partial \boldsymbol{x}}{\partial v}, \quad \boldsymbol{e}_w = \frac{1}{h_w}\frac{\partial \boldsymbol{x}}{\partial w},$$

where

$$h_u = \left|\frac{\partial \boldsymbol{x}}{\partial u}\right|, \quad h_v = \left|\frac{\partial \boldsymbol{x}}{\partial v}\right|, \quad h_w = \left|\frac{\partial \boldsymbol{x}}{\partial w}\right|$$

are called the scale factors.

Remark 2.6.3. Geometrically, the scale factors tell you how much an infinitesimal region in (u, v, w) coordinates is stretched at different points in real space. This is critical to account for during surface and volume integrals

Remark 2.6.4. It is relevant here to repeat the two coordinate basis from before:

Cylindrical polars	**Spherical polars**
Conversion formulae	
$x_1 = \rho\cos\varphi$ $x_2 = \rho\sin\varphi$ $x_3 = z$	$x_1 = r\sin\theta\cos\varphi$ $x_2 = r\sin\theta\sin\varphi$ $x_3 = r\cos\theta$
Basis vectors	
$\mathbf{e}_\rho = (\cos\varphi, \sin\varphi, 0)$ $\mathbf{e}_\varphi = (-\sin\varphi, \cos\varphi, 0)$ $\mathbf{e}_z = (0, 0, 1)$	$\mathbf{e}_r = (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)$ $\mathbf{e}_\varphi = (-\sin\varphi, \cos\varphi, 0)$ $\mathbf{e}_\theta = (\cos\theta\cos\varphi, \cos\theta\sin\varphi, -\sin\theta)$

Chapter 3

Differential Operators

3.1 Differentiating Scalar Fields

Definition 3.1.1 (Gradient of a Scalar Field)**.** The gradient of a scalar field $f : \mathbb{R}^3 \to \mathbb{R}$ is the operator

$$\nabla f|_{\boldsymbol{x}} = \lim_{\mathrm{d}(S)\to 0} \frac{1}{|V|} \oint_S f\hat{\boldsymbol{n}} dS,$$

where S is a closed surface around $\boldsymbol{x}$ with outward normal $\hat{\boldsymbol{n}}$, enclosed volume $|V|$, and diameter $\mathrm{d}(S) = \sup_{\boldsymbol{x},\boldsymbol{y}\in S} |\boldsymbol{x} - \boldsymbol{y}|$.

Remark 3.1.2. We note that ∇f is a vector field $\mathbb{R}^3 \to \mathbb{R}^3$ and our intuition is that it points in the direction of the fastest increase of f.

Proposition 3.1.3. In Cartesian coordinates,

$$\boldsymbol{\nabla} f = \frac{\partial f}{\partial x}\boldsymbol{e}_1 + \frac{\partial f}{\partial y}\boldsymbol{e}_2 + \frac{\partial f}{\partial z}\boldsymbol{e}_3.$$

This tells us that ∇f is a differential operator, acting on a scalar field and returning a vector field. We see a similar formula from Calculus I but the integral we have defined here should work independent of any particular choice of coordinates. To prove our proposition we use the following lemma:

Lemma 3.1.4 (Mean Value Theorem for Double Integrals)**.** If a scalar field $f : \mathbb{R}^2 \to \mathbb{R}$ is continuous on a closed, bounded and connected domain $A \subset \mathbb{R}^2$ then there exists $\boldsymbol{x}_* \in A$ such that

$$\int_A f dS = |A| f(\boldsymbol{x}_*)$$

This formula is analagous for the 1-d result of mean value theorem depicted in the figure below. In 1-d we have that if $f(x)$ is continuous on $[a, b]$ then there exists $x_* \in [a, b]$ such that $\int_a^b f(x)dx = (b - a)f(x_*)$.

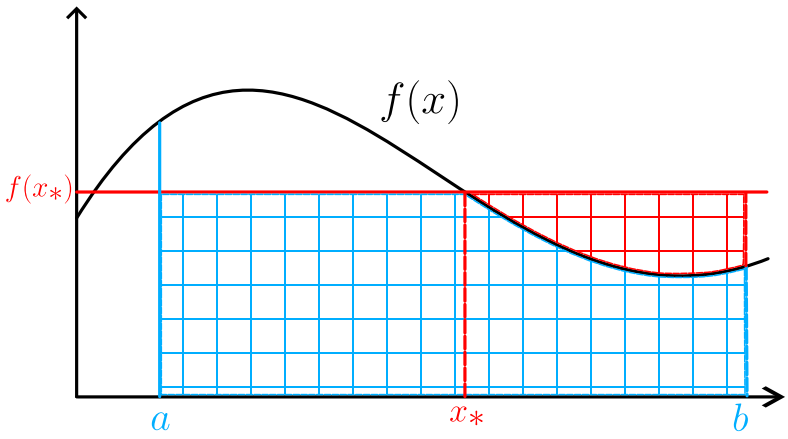


Figure 3.1: 1D representation of Mean Value Theorem

Lemma 3.1.4 says that there is some particular point $\boldsymbol{x}_*$ where f attains its average value, $f(x_*) = \frac{1}{|A|}\int_A f dS$.

Proposition 3.1.5. In orthogonal curvilinear coordinates (u, v, w), the gradient has the expression

$$\boldsymbol{\nabla} f = \frac{1}{h_u}\frac{\partial f}{\partial u}\boldsymbol{e}_u + \frac{1}{h_v}\frac{\partial f}{\partial v}\boldsymbol{e}_v + \frac{1}{h_w}\frac{\partial f}{\partial w}\boldsymbol{e}_w.$$

Example 3.1.6 (Computing ∇z in both Cartesian and Spherical Coordinates)**.** In Cartesians, we have

$$\boldsymbol{\nabla}(z) = \frac{\partial z}{\partial x}\boldsymbol{e}_1 + \frac{\partial z}{\partial y}\boldsymbol{e}_2 + \frac{\partial z}{\partial z}\boldsymbol{e}_3 = \boldsymbol{e}_3.$$

In spherical coordinates (r, θ, ϕ), recall that $z = r\cos\theta$. So Proposition 3.2 gives

$$\begin{aligned}\boldsymbol{\nabla}(r\cos\theta) &= \frac{1}{h_r}\frac{\partial}{\partial r}(r\cos\theta)\boldsymbol{e}_r + \frac{1}{h_\theta}\frac{\partial}{\partial\theta}(r\cos\theta)\boldsymbol{e}_\theta + \frac{1}{h_\phi}\frac{\partial}{\partial\phi}(r\cos\theta)\boldsymbol{e}_\phi \\ &= \frac{\partial}{\partial r}(r\cos\theta)\boldsymbol{e}_r + \frac{1}{r}\frac{\partial}{\partial\theta}(r\cos\theta)\boldsymbol{e}_\theta \\ &= \cos\theta\boldsymbol{e}_r - \sin\theta\boldsymbol{e}_\theta.\end{aligned}$$

We recognise this as $\boldsymbol{e}_3$ again.

Remark 3.1.7. Observe that our coordinate formula, and hence $\boldsymbol{\nabla} f$, extends to scalar fields on any $\mathbb{R}^n$, and giving always a vector in $\mathbb{R}^n$. Very high-dimensional vectors are commonly used in machine learning algorithms.

3.2 Differentiating Vector Fields

There are two standard ways to differentiate a vector field $\mathbf{f} : \mathbb{R}^3 \to \mathbb{R}^3$:

– The divergence of $\boldsymbol{f}$ is the scalar field

$$\boldsymbol{\nabla}\cdot\boldsymbol{f}|_{\boldsymbol{x}} = \lim_{\mathrm{d}(S)\to 0}\frac{1}{|V|}\oint_S \hat{\boldsymbol{n}}\cdot\boldsymbol{f}dS.$$

– The curl of $\boldsymbol{f}$ is the vector field

$$\boldsymbol{\nabla}\times\boldsymbol{f}|_{\boldsymbol{x}} = \lim_{\mathrm{d}(S)\to 0}\frac{1}{|V|}\oint_S \hat{\boldsymbol{n}}\times\boldsymbol{f}dS.$$